PATIENT DEMOGRAPHICS

- Patient name
- DOB
- Sex
- Hospital / NHS number

RADIOGRAPH DETAILS

- **Date and Time**

- **Orientation:**
  - Erect / Supine:
    - Erect CXR to look for pneumoperitoneum
    - Patient must be upright for at least 15 minutes beforehand
  - PA / AP
    - Assume PA unless stated
    - AP magnifies heart and mediastinum

- **Adequacy:**
  - Should see whole thorax, including:
    - Lung apices
    - Costophrenic angles
  - Inspiration:
    - 6th rib should intersect right hemidiaphragm in mid clavicular line
  - Exposure:
    - Able to just visualise thoracic vertebrae
  - Rotation:
    - Spinous process should be equidistant between medial heads of clavicles

RADIOGRAPH INTERPRETATION

A	B	C	D	E	S
Airway	Breathing	Circulation	Diaphragm	Extra	Soft Tissue

Airway (Trachea) + Mediastinum

Trachea:
- Should be central
- Will normally deviate to right as goes past aortic knuckle.

Trachea pulled towards lesion:
- Lung / lobe collapse
- Fibrosis
- Pneumonectomy

Trachea away from lesion:
- Tension pneumothorax
- Large pleural effusion
- Space occupying lesion

Mediastinum (great vessels, trachea, oesophagus, thymus, lymph nodes, phrenic nerves, vagus nerve):
- Should be central
- < 8cm width PA film

Hila:
- Left hilum should be 2cm higher than Right hilum

Mediastinal widening:
- Potential lymphadenopathy / aortic aneurysm / aortic dissection / hiatus hernia

Displaced Mediastinum / Hila:
- Lung / lobe collapse
- Fibrosis
- Pneumonectomy
- Tension pneumothorax
- Large pleural effusion
- Space occupying lesion

Enlarged Hila:
- Potential malignancy / Sacrdosis (bilateral) / Pulmonary hypertension

Breathing (Lungs)

Lung Markings:
- Lung marking should extend to edge of thorax
- Lung markings should be similiar denisty throughout
- Fissures between lobes not usually visable

Lung markings not to edge of thorax:
- Pleural space Darker:
  - Pneumothorax
- Pleural space Whiter:
  - Effusion / Haemothorax / Empyema

Areas of greater density (whiter):
- Diffuse:
  - Infection
  - Atelectasis
  - Pulmonary oedema (bat wing appearance)
- Localised / well defined:
  - Masses / Infiltrates
  - Calcifications

Areas of less dense (darker) lung markings:
- Emphysematous changes / Bullae

Prominent fissures:
- Pulmonary oedema (fluid in fissure)

Circulation (Heart)

Heart borders (contours):
- Should see clear heart borders down to diaphragm

CTR (cardiothoracic ratio):
- <50% on PA film

Heart border obscured:
- Right side - consider right middle lobe consolidation
- Left side - consider lingular consolidation

CTR > 50%:
- PA film suggests cardiomegaly (possible Heart Failure)
- Unable to comment on AP film as heart falsely enlarged

Enlarged Hila:
- Potential malignancy / Sacrdosis (bilateral) / Pulmonary hypertension

Diaphragm

Hemidiaphragms:
- Due to liver Right should be higher than Left
- Should be dome shaped
- Gastric bubble under left

Cardiophrenic & Costophrenic angles:
- Should be clear and sharp
- Cardiophrenic angle may contain fat density

Raised Hemidiaphragm:
- Phrenic nerve damage/palsy:
  - Can be caused by masses in the upper lobes compressing the nerve as it descends
- Lung disease causing volume loss:
  - Fibrosis
  - Lobectomy
  - Atelectasis

Flattened Hemidiaphragm (>6th rib intersecting diaphragm)**:**
- Overexpanded chest:
  - Obstructive airways disease e.g. COPD
  - Excessive inspiration

Blunted Costophrenic angle
- With meniscus suggests pleural effusion
- Without meniscus suggests consolidation

Subdiaphragmatic Air on Erect CXR:
- Pneumoperitoneum:
  - Suggests perforation of intra-abdo viscus

Extras

Artifacts:
- Medical lines:
  - NG tubes, PICC line, Central line, Chest drain
- Implants:
  - Pacemaker
- Surgery:
  - Heart valves
  - Sternotomy wires

NG Tube:
- Check correct position

Soft Tissue + Bones

Soft Tissues:
- Should be reasonably uniform in density
- Breast shadows

Bones:
- Should have clear outlines
- Equal on both sides
- Check Ribs, Clavicles, Humerus

Soft-tissue Opacity Neck:
- Potential mass, lymphadenopathy, goitre

Uneven / increased density breast shadows:
- Has there been any breast surgery

Subcutaneous air (gas) :
- Surgical emphysema

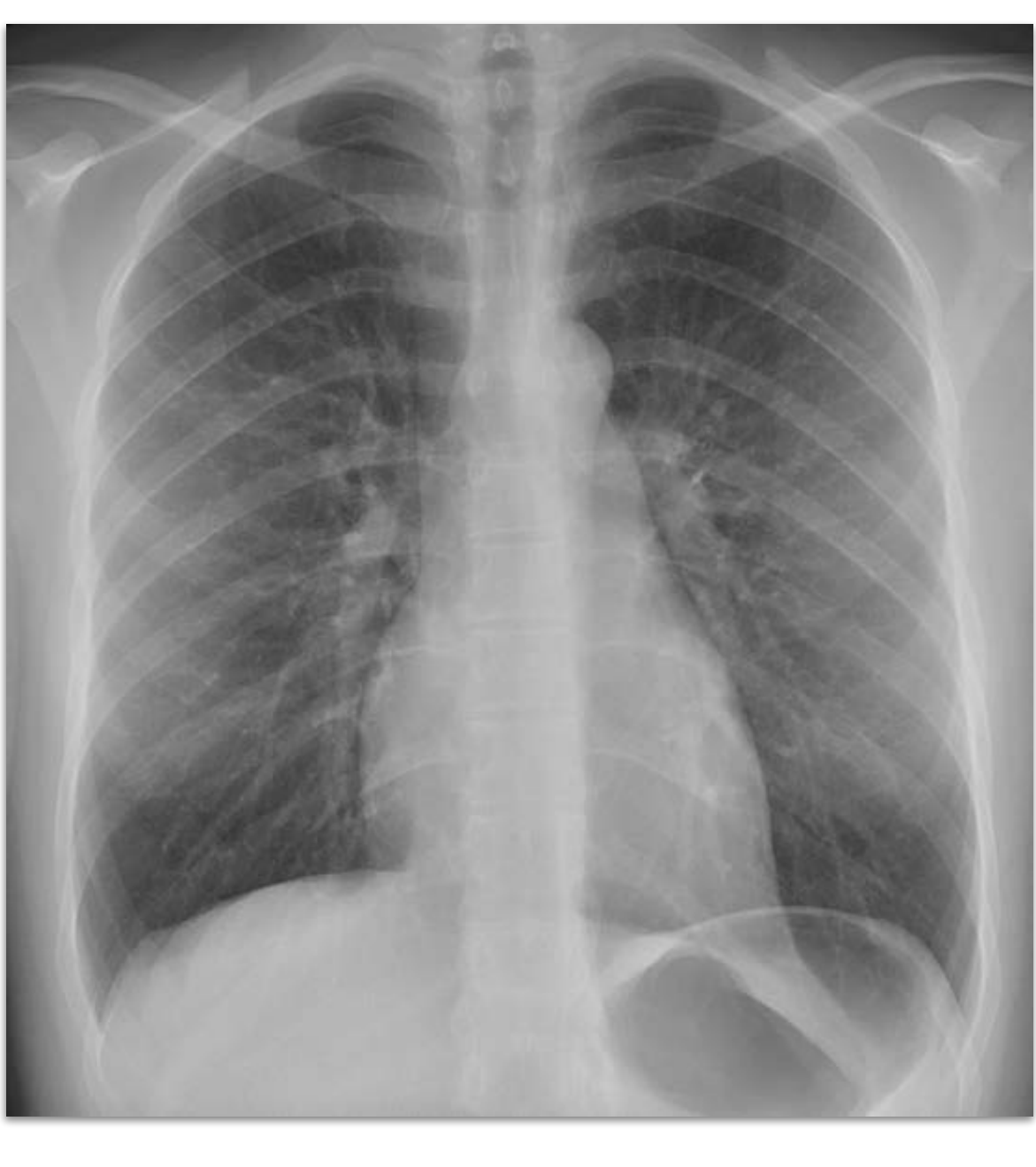
Bone irregularities :
- Rib / Clavicle / Humeral fractures
- Shoulder dislocations
- Primary / Metastatic lesion: Lytic (darker) Sclerotic (whiter)

Final Points

- **Comparison:**
  - If available compare to previous X-ray
- **Focus:**
  - Examine whole X-ray, don't focus on obvious abnormality
- **Patient:**
  - Always interpret X-ray with clinical picture in mind

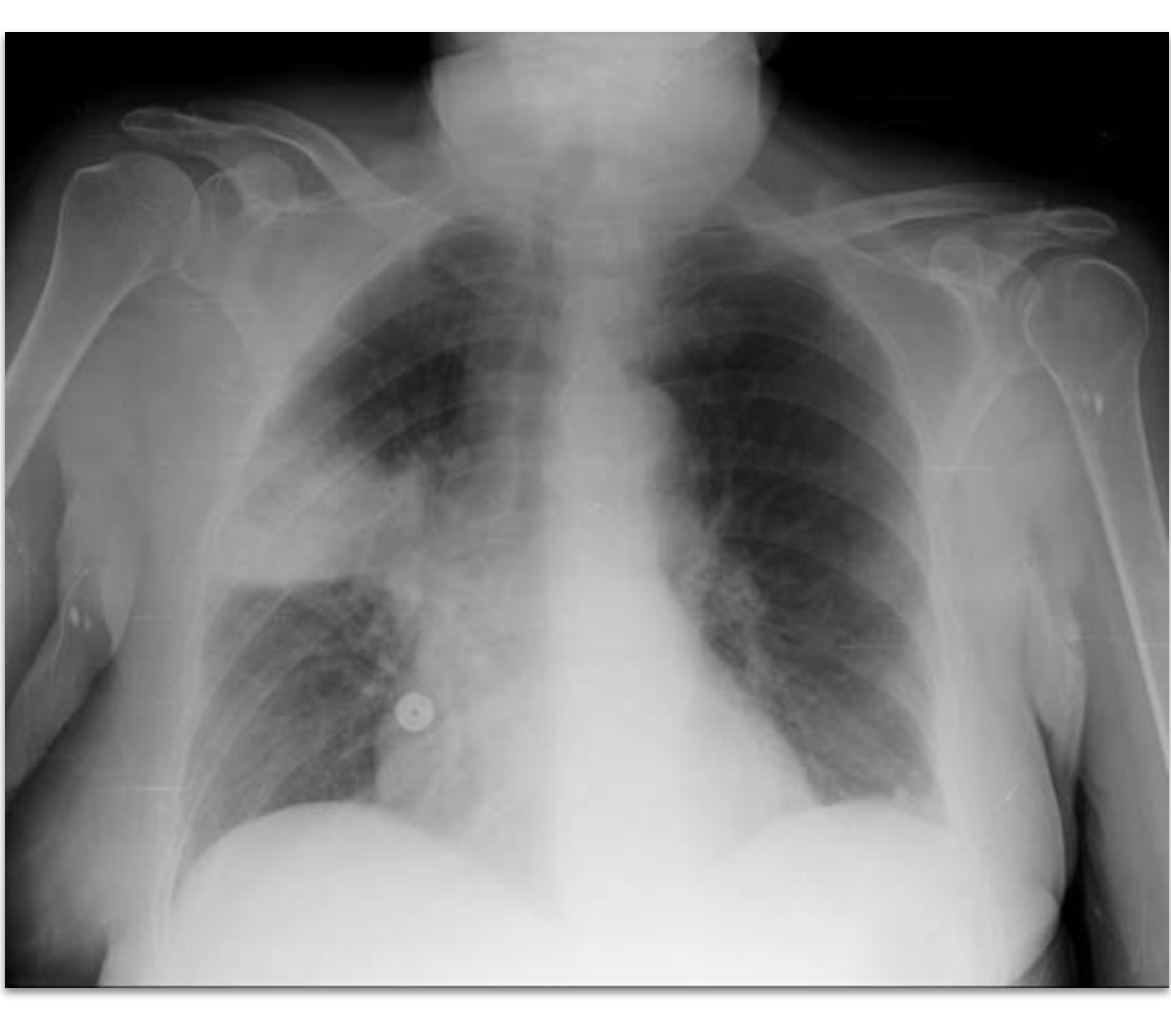
NORMAL CXR

Normal PA CXR

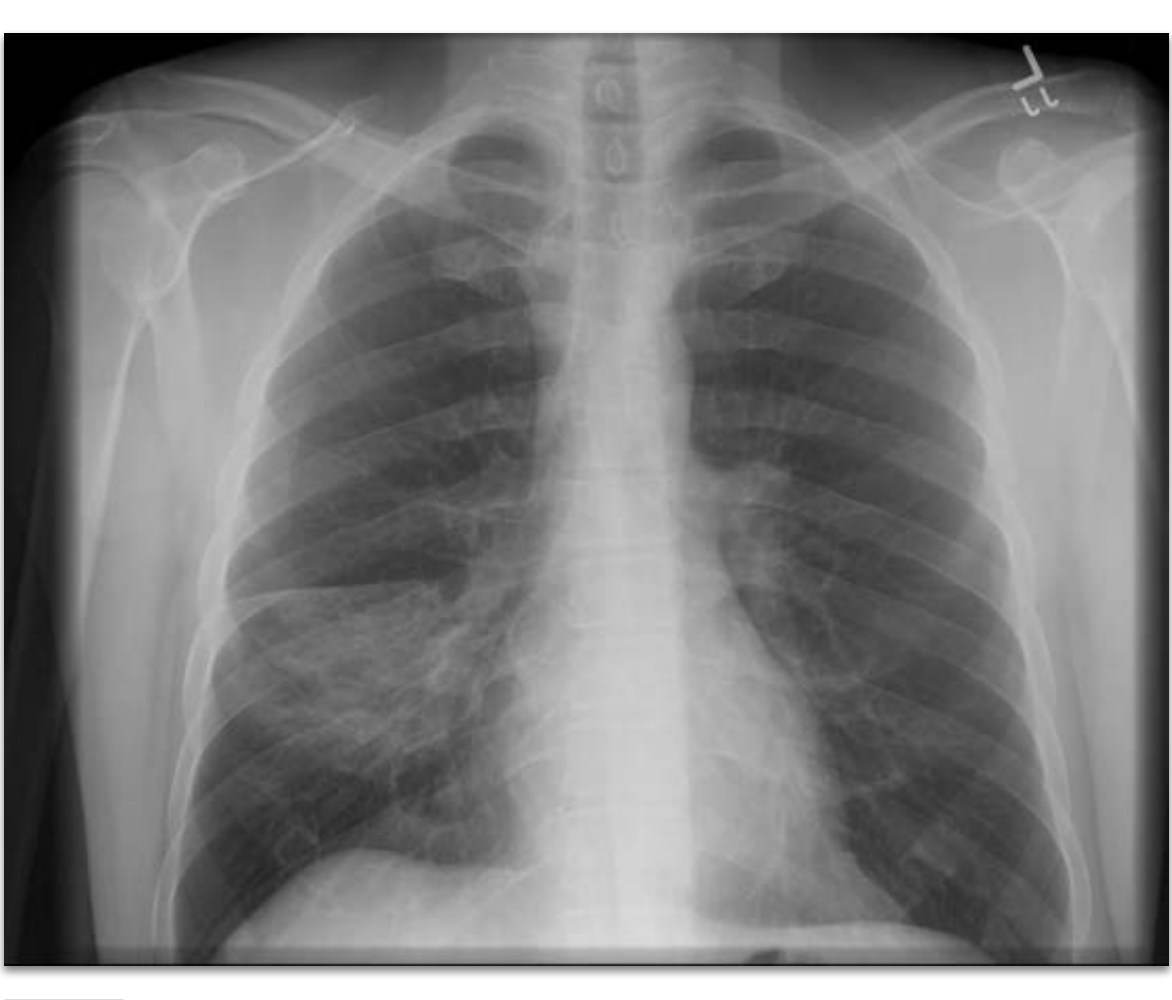


COMMON FINDINGS

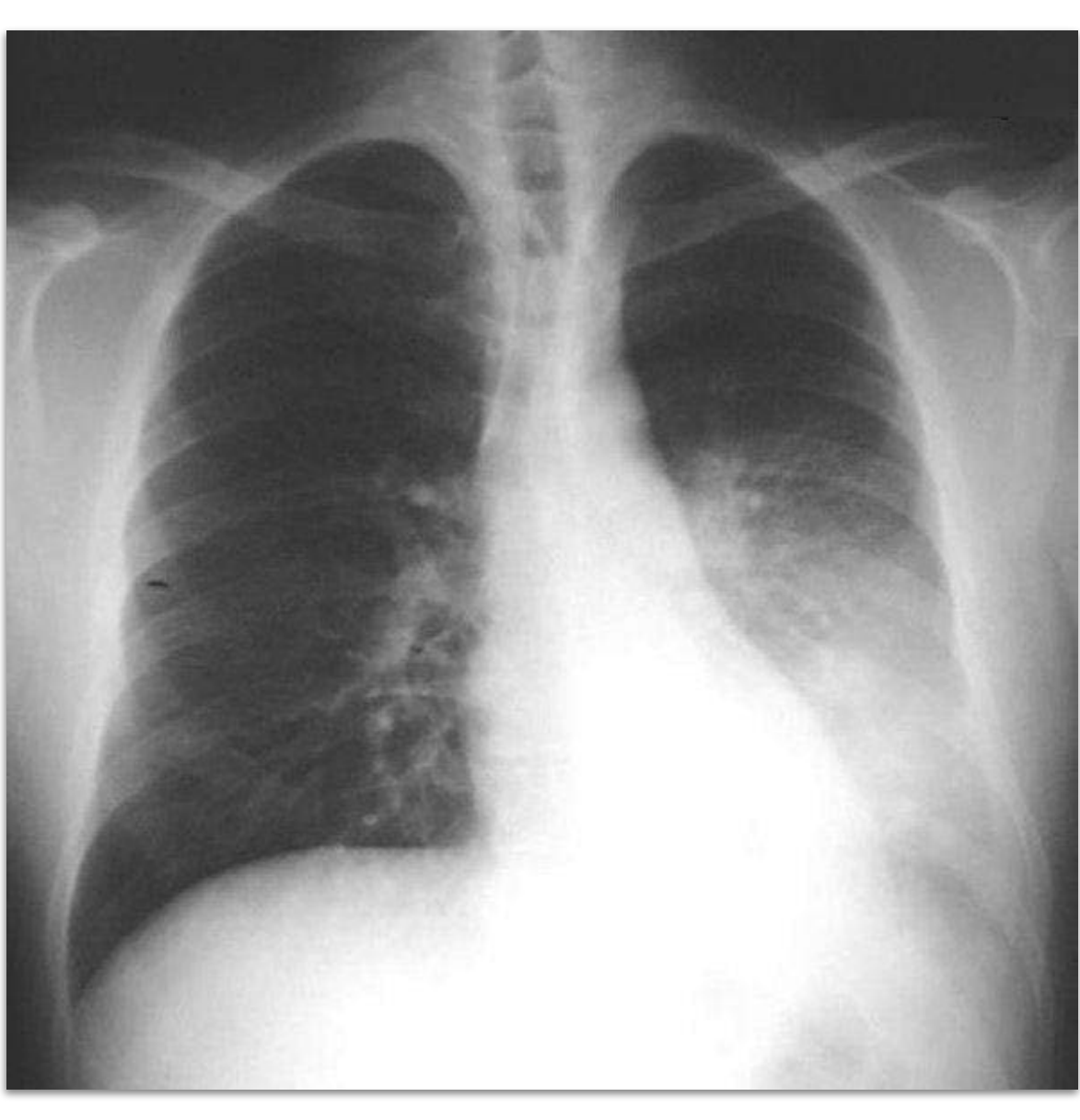
Consolidation - Right Upper Lobe Pneumonia
- Opacity right upper lobe
- Above horizontal fissure



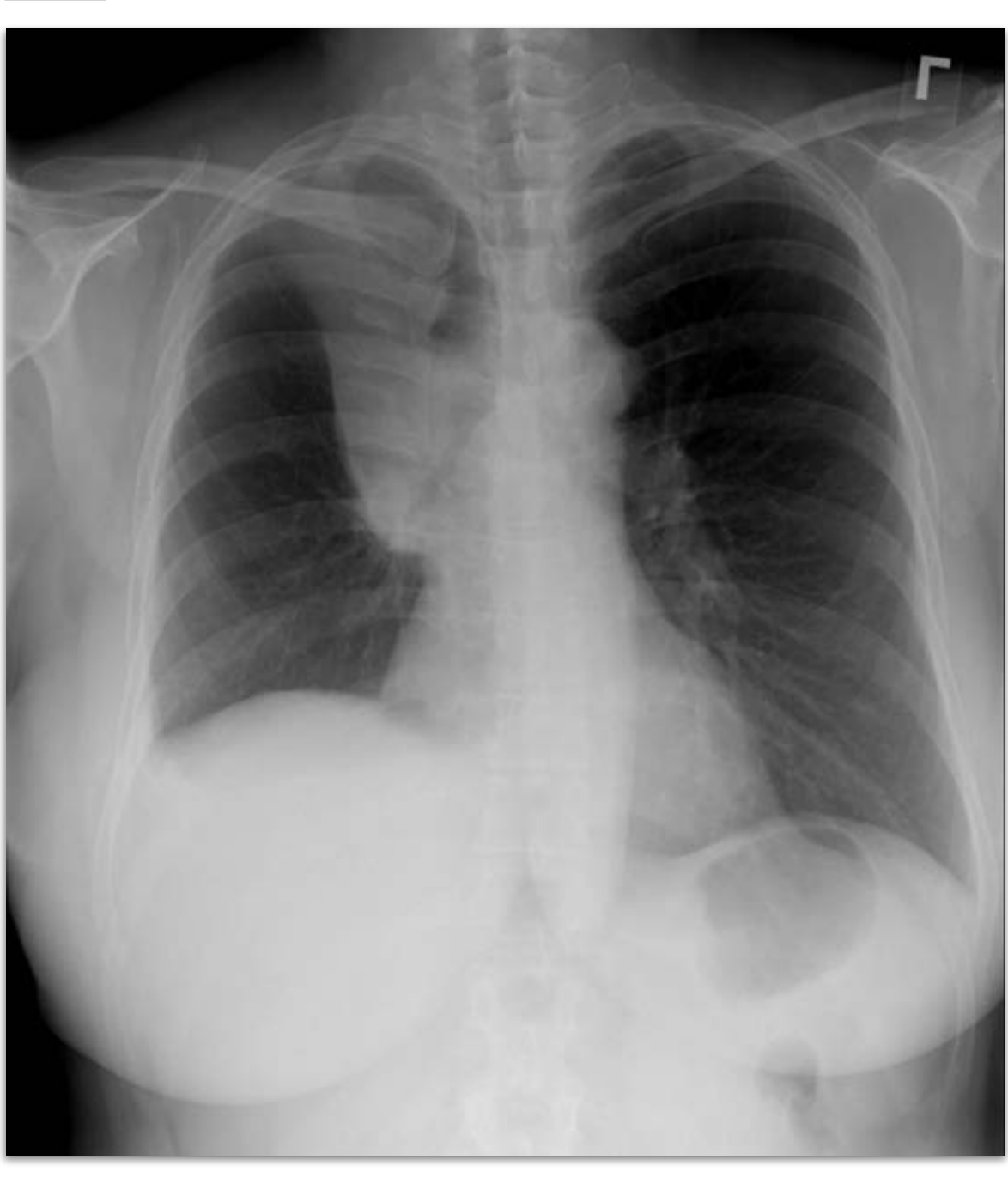
Consolidation - Right Middle Lobe Pneumonia
- Opacity Right middle lobe
- Obscuring right heart border and below horizontal fissure



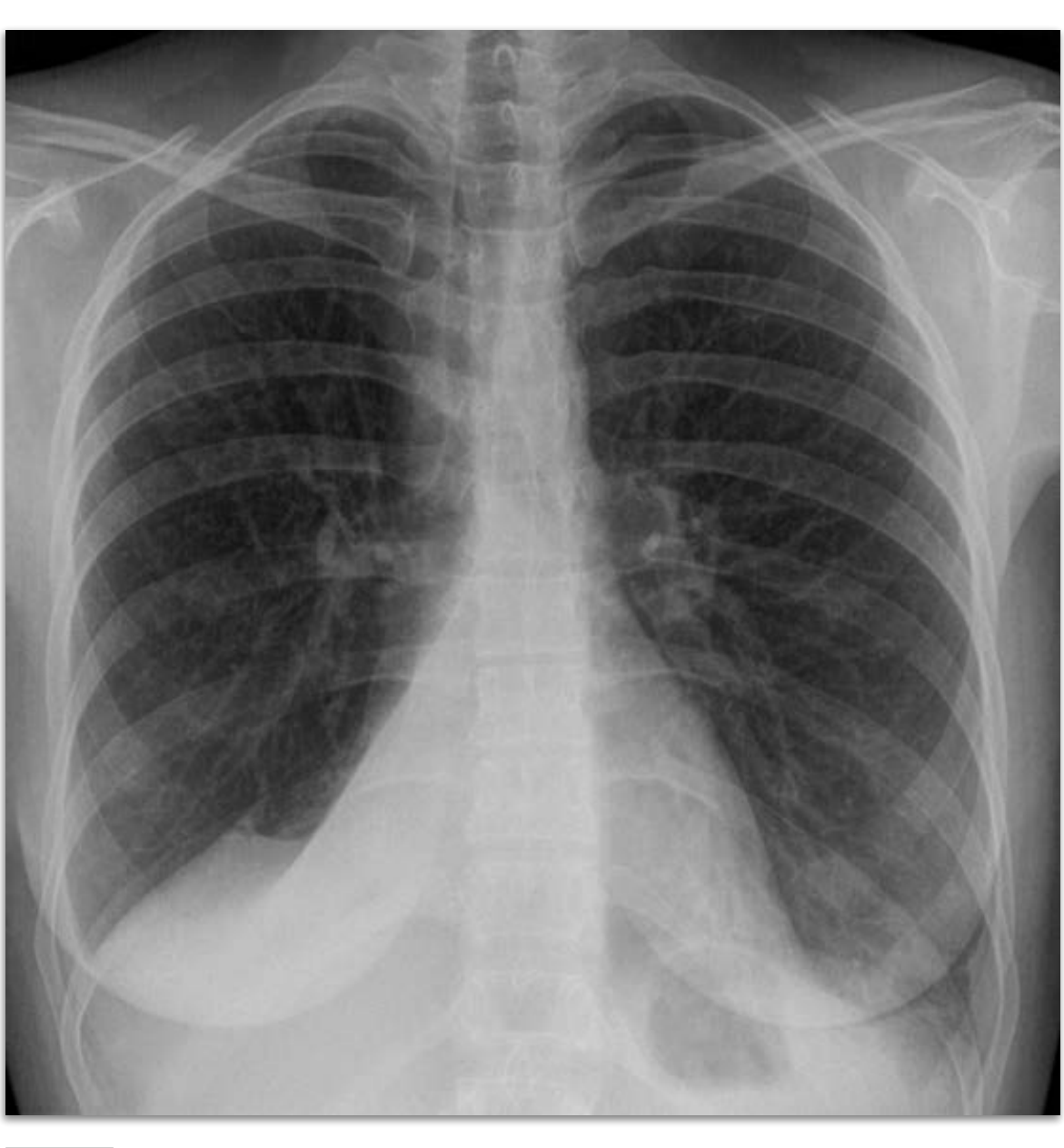
Consolidation - Left Lower Lobe Pneumonia
- Opacity Left lower lobe
- Obscuring left hemidiaphragm and costophrenic angle



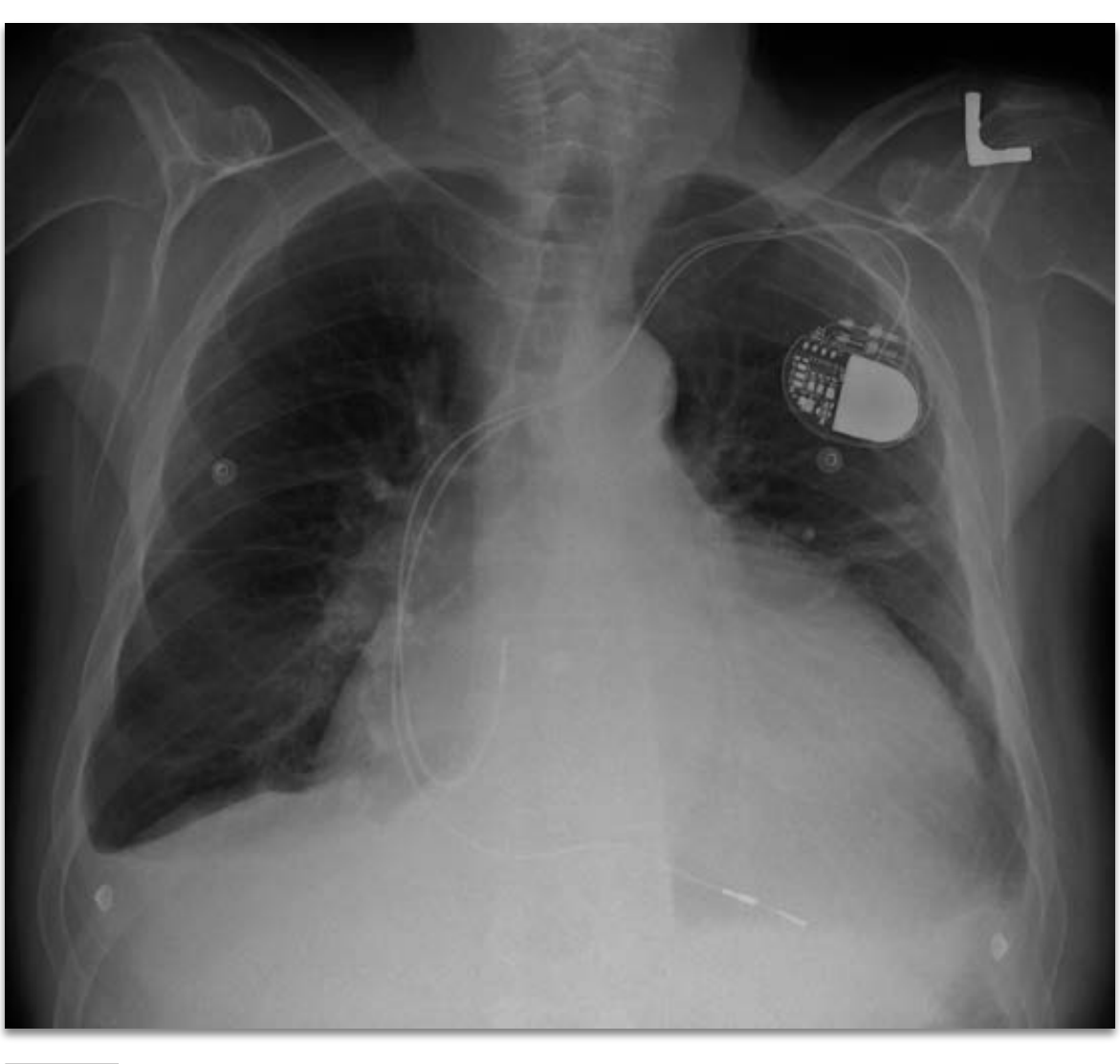
Right Upper Lobe collapse
- Right upper lobe opacity
- Raised right hemidiaphragm
- Distorted horizontal fissure



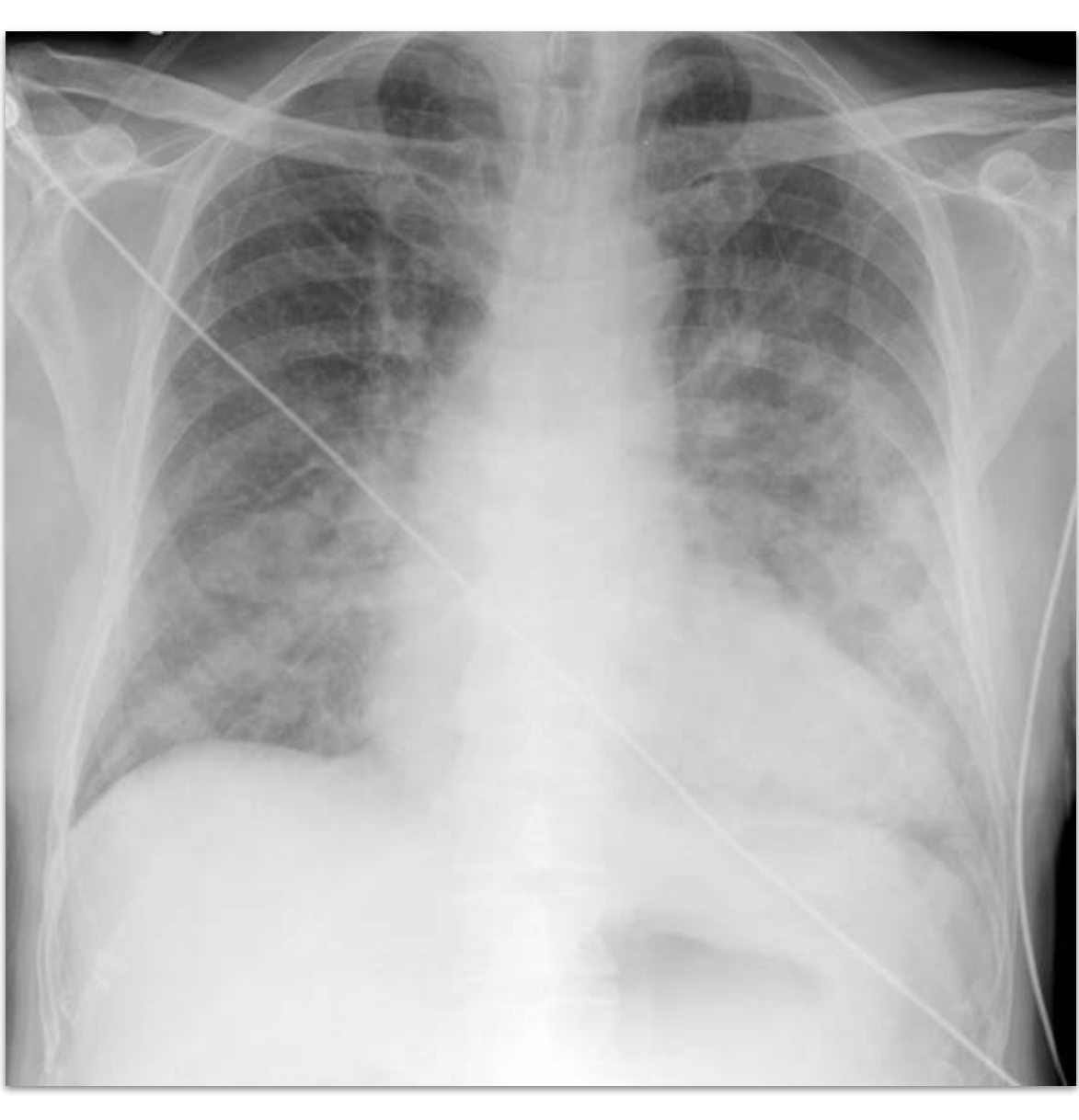
Right Lower Lobe collapse
- OpacityRight lower lobe - Sail sign



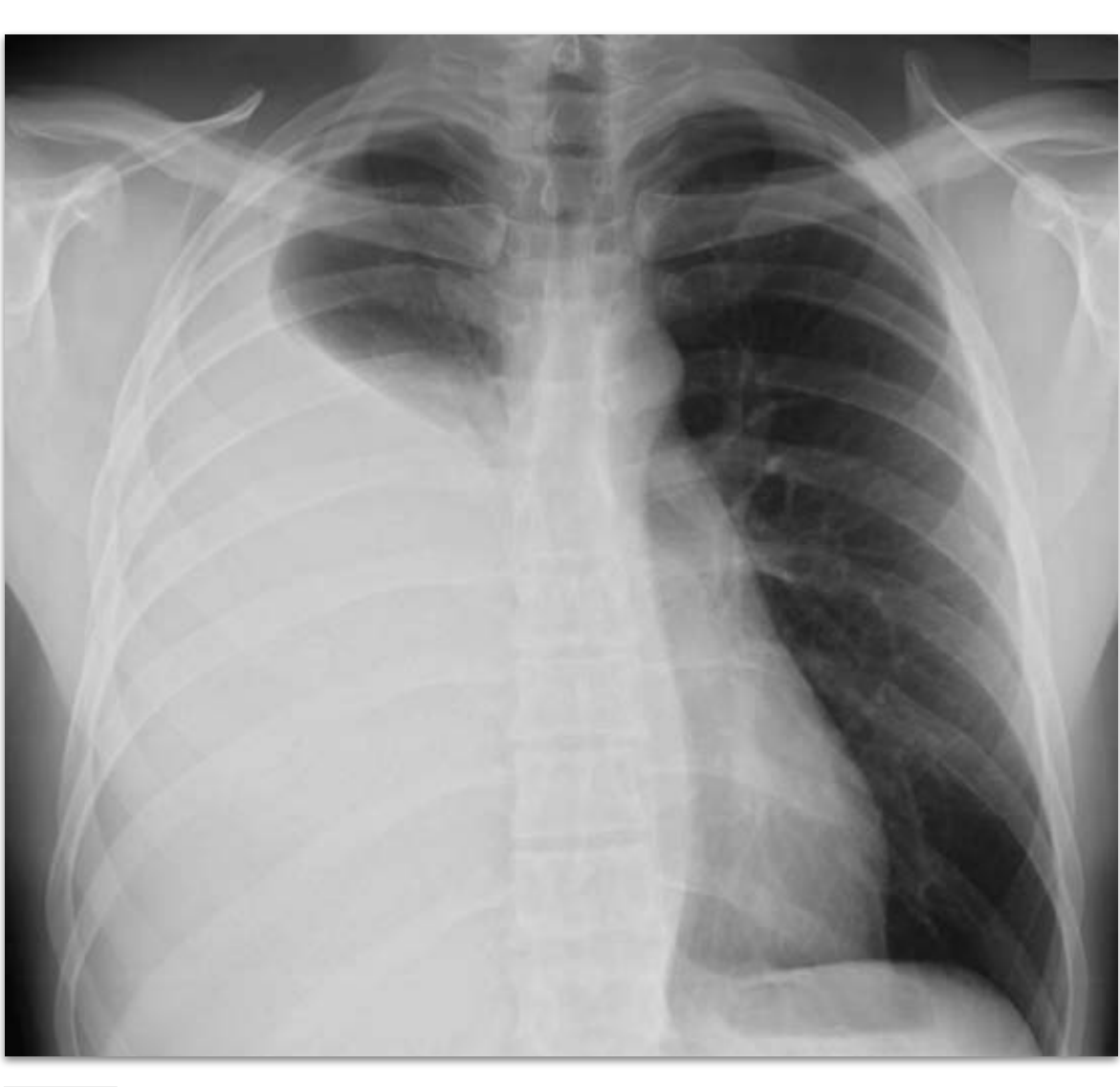
Cardiomegaly / Pleural effusions:
- Heart > 50% CTR + Pacemaker in situ
- Bilateral pleural effusions



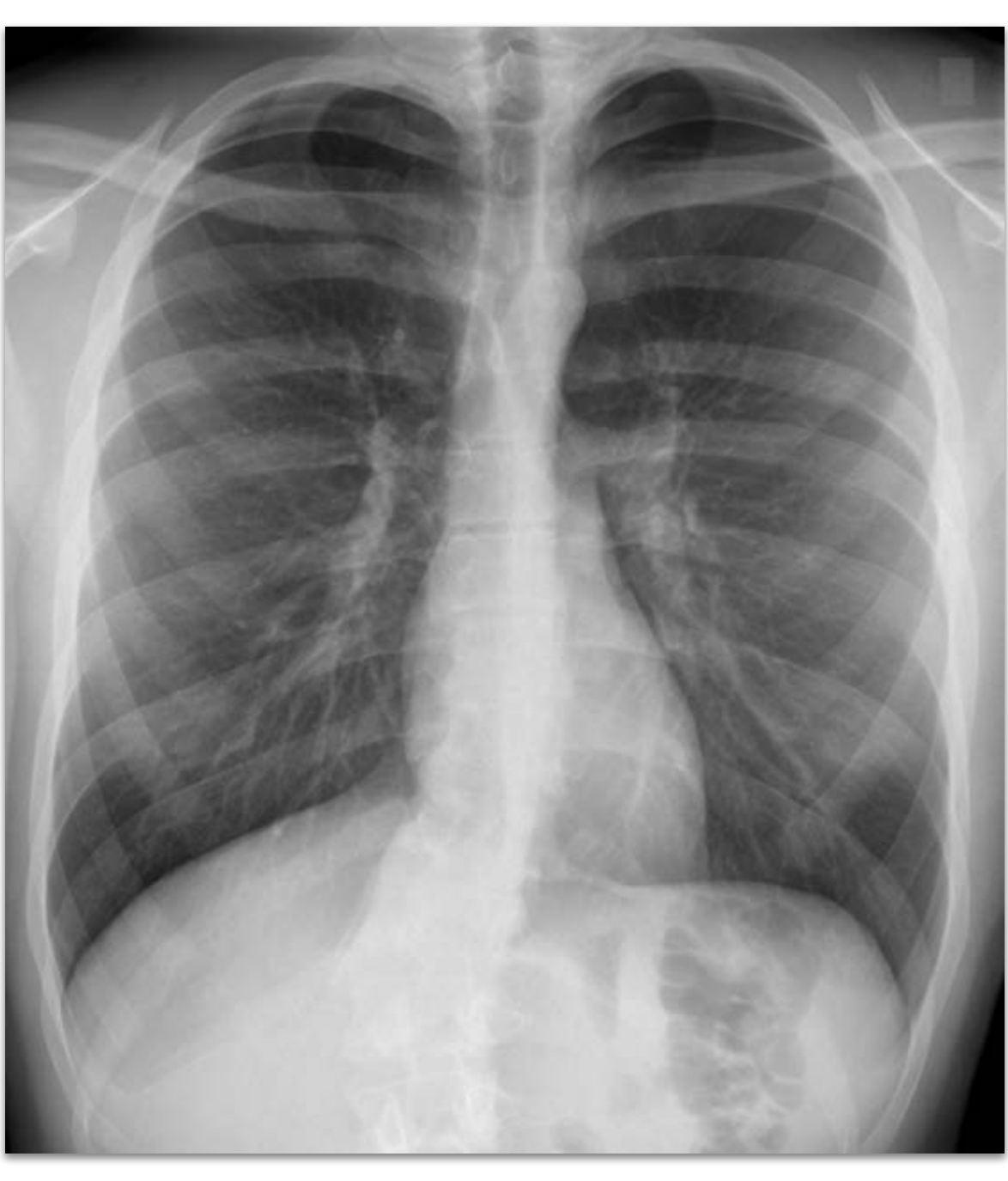
Pulmonary Oedema:
- Widespread diffuse opacity in lung fields
- Spreading from hilum



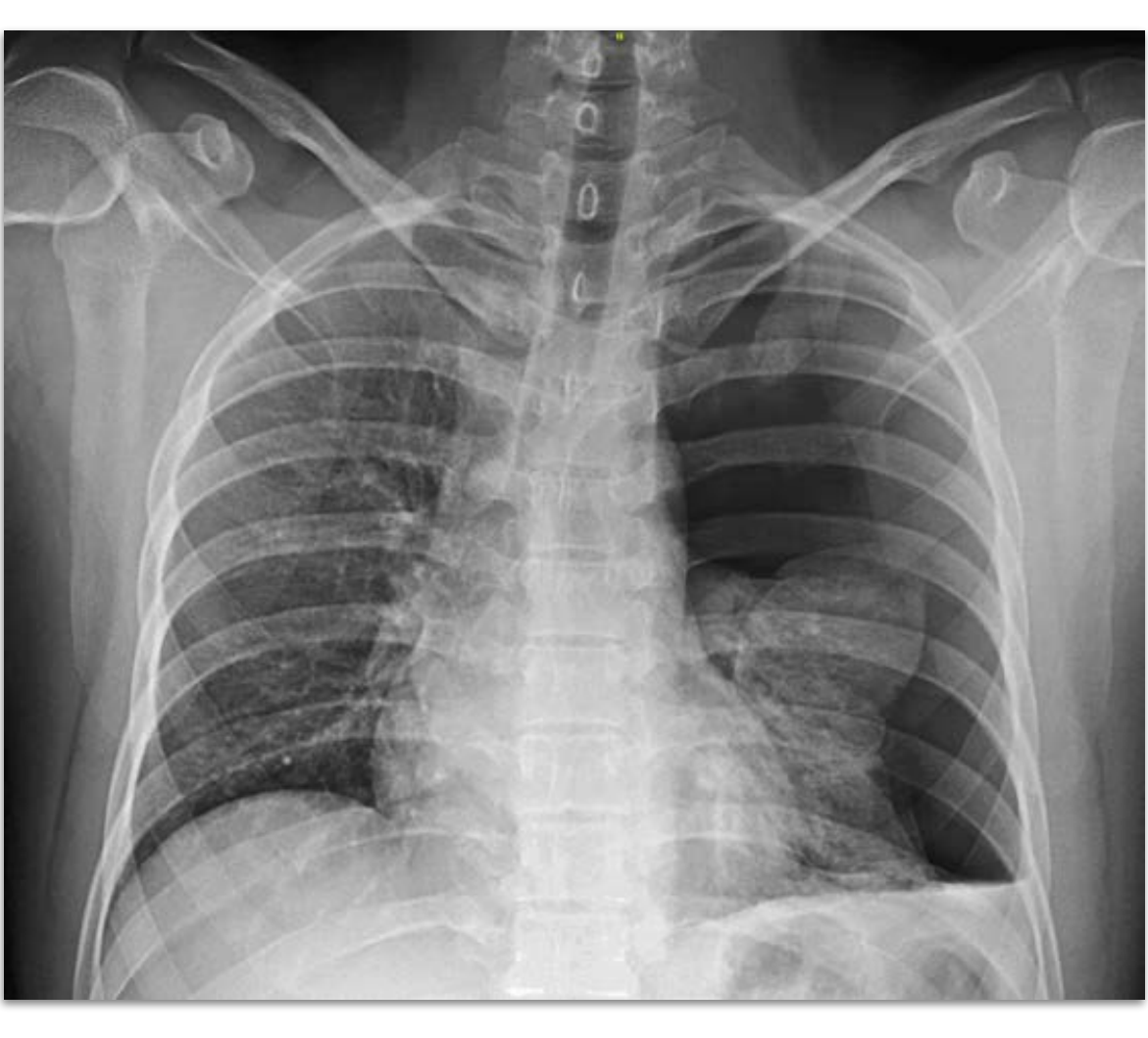
Pleural Effusion:
- Large right sided opacity with meniscal line on lateral edge



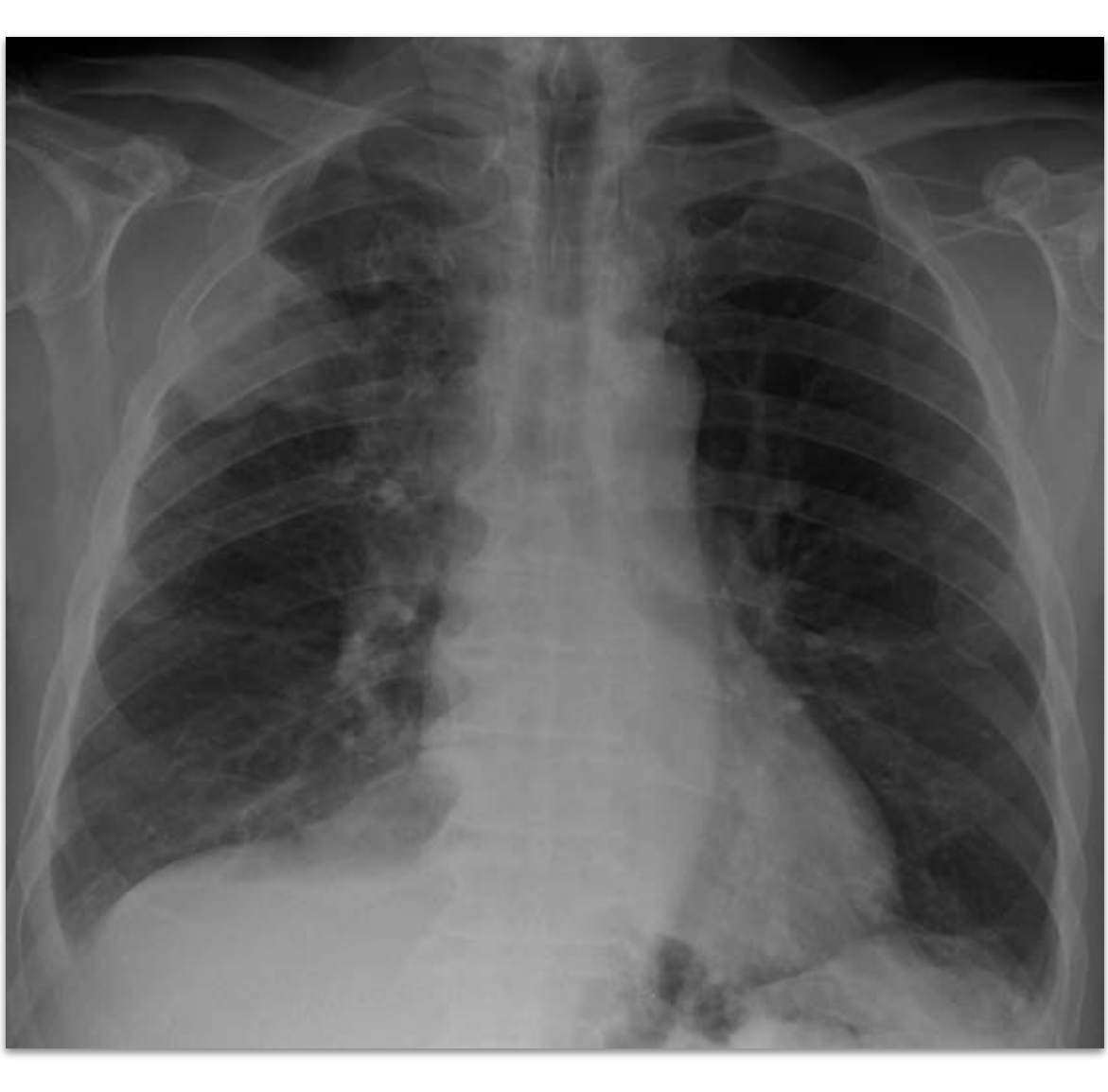
Simple Small Left Pneumothorax:
- No lung markings left upper zone



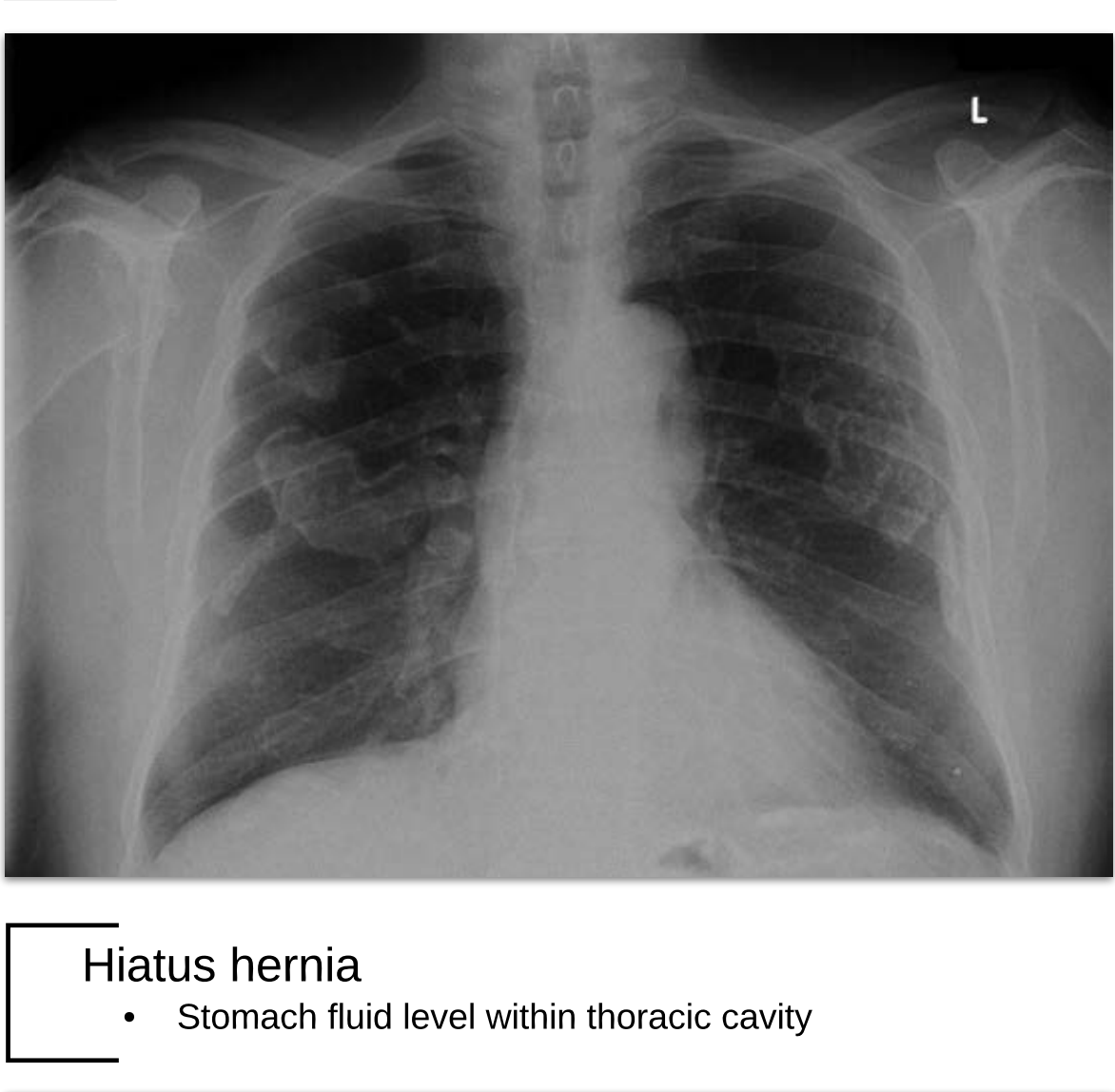
Tension Pneumothorax - Left side:
- Loss lung markings left side
- Right sided shift mediastinum

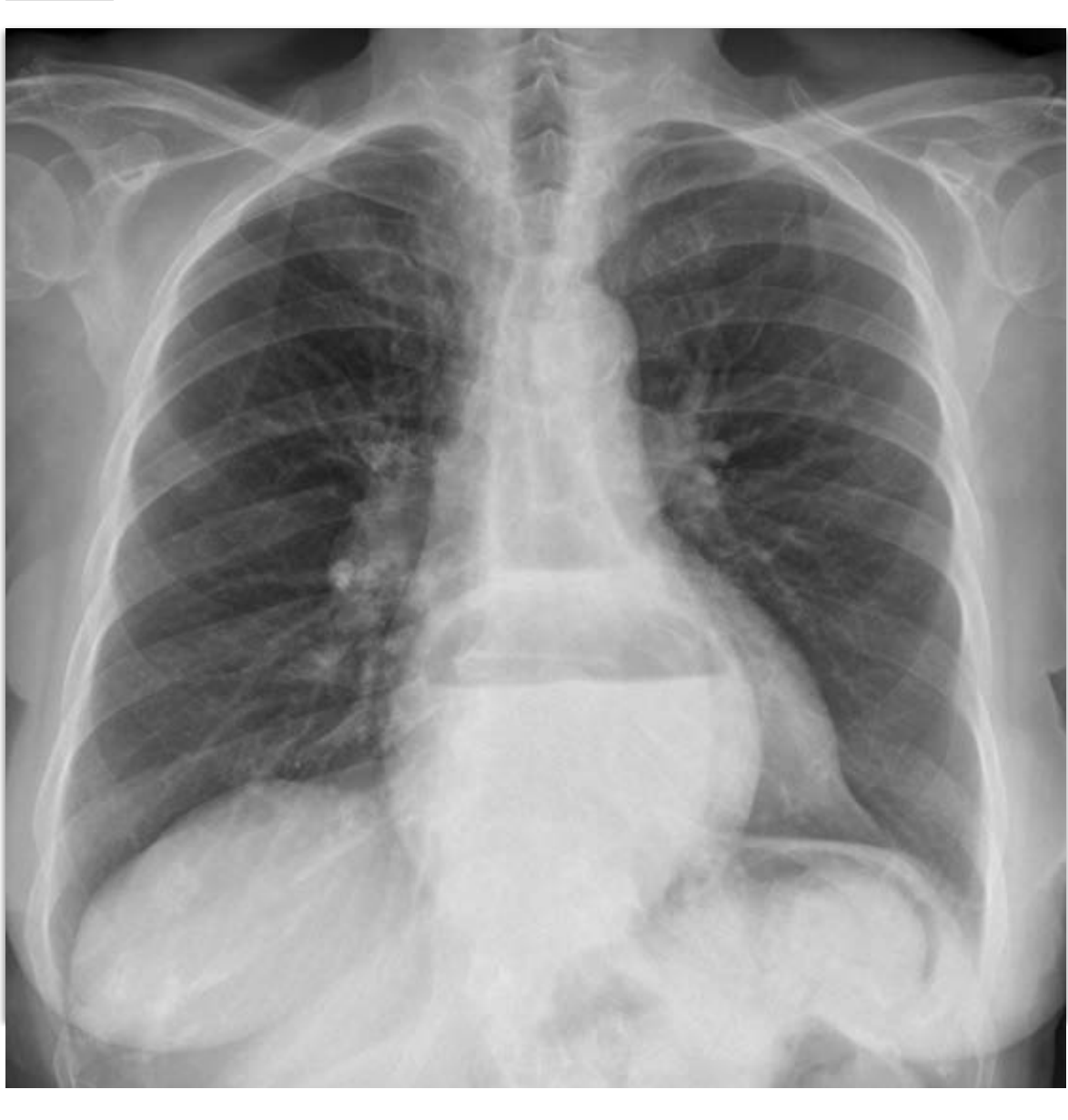


Lung mass:
- Right sided well demarcated lung opacity
- Hilar lymphadenopathy



Plural plaques:
- Well demarcated opacities throughout both lung fields
- High density of outline of opacities suggesting calcification



Hiatus hernia
- Stomach fluid level within thoracic cavity